

Log transforming the response

Multiplicative increments

- The **logarithmic transformation** of the response is a common transformation when the effect of the predictor on the response is thought to be multiplicative
 - A change of a unit in x is associated with a constant **percentage** change in y
- Many processes benefit from log transformation:
 - Growth is proportional do previous size
 - Any multiplicative process

$$y_i \sim N(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta x_i$$



$$\Delta y \approx \beta \Delta x$$

$$\log(y_i) \sim N(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta x_i$$



$$\% \Delta y \approx (100 \times \beta) \Delta x$$

Biomass by diameter

Example of non-linear relation

Option 1: log-transform y

```
df = data.frame(diameter, biomass)
fit = lm(log(biomass) ~ diameter,
         data = df)
```

Option A: Assign y a log-normal likelihood

```
rt_fit = quap(alist(
  biomass ~ lognormal(mu, sigma),
  mu <- a + b*diameter),
  data = df)
```

